Redbull CCTV — Architecture Design

1. Problem, scope

Today nobody can connect a campaign click to the order it produced: ONDC sees the clicks, and Buyer Apps see the orders. So Marketing cannot say which campaign worked, which affiliate sends real buyers, or whether a drop came from the offer, the audience or a broken page.

The platform closes that gap. It routes campaign traffic, receives what happens on the Buyer App, matches orders back to clicks, and tells Marketing what changed and what to do about it.

Scope of Phase 1

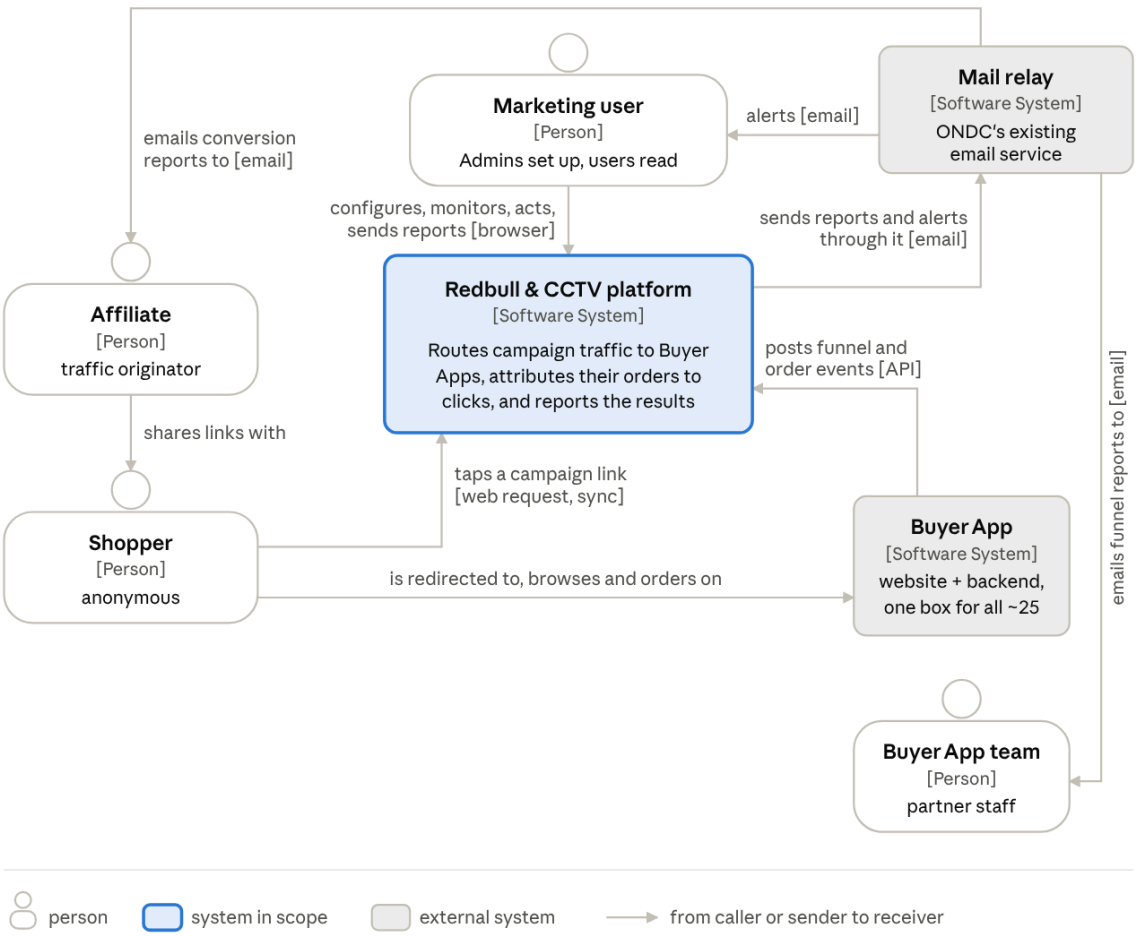
In Phase 1	**Not in Phase 1**
Campaign setup and traffic routing across Buyer Apps	Shoppers without the app installed, and app-store redirects
Funnel and order events from Buyer App websites, posted by their backends	Journeys across devices or browsers
Order attribution within attribution windows	Insight text written by a language model
Category match and ROI reporting	Reversing attribution when an order is cancelled or returned
Rule-based flags and a decision queue	Live postbacks to affiliates
Affiliate reports emailed from the platform	Mobile apps and SDKs
Configuration versions, audit history and rollback	

2. Actors and system boundary

Seven actors surround the platform: five kinds of person and two software systems. Only the Buyer App calls the platform system to system, and the platform calls only the mail relay. Everyone else uses the UI, passes through a redirect, or receives email. The diagram draws Marketing's two roles as one box.

System context: only Buyer Apps connect to the platform system to system

C4 level 1 · scope: the Redbull & CCTV platform, Phase 1



Brackets on arrows give the interaction style; technology is still to be chosen.

Who they are and what they exchange

Actor	Gives the platform	Gets from the platform
Marketing admin	Buyer App and campaign setup, thresholds and targets, each change with a reason; rollbacks;	Validation results, versions and audit history, plus everything a general user gets
Marketing general user	Flag dismissals with reasons; report sends	Dashboards, funnels, the daily affiliate view and the decision queue; outage alerts by email
Shopper	One tap on a campaign link	A redirect to one Buyer App, carrying the click id. Nothing about the person is kept

Affiliate	Its own ids on every tap	One fixed link per campaign; conversion reports by email that return its ids
Buyer App	Funnel and order events from its backend; test events before go-live	Shoppers carrying a click id; an accept or reject answer, with the reason, for every event
Buyer App team	The one-time integration against the published schema	The schema and credentials at onboarding; its funnel report by email, as its tier allows
Mail relay	Delivery of each mail	

3. Requirements

A. Configuration & Registry — S1

The PRD says S1 owns marketing configuration and Buyer App setup, including campaign configuration, Buyer App registry, event contract, versions, audit and rollback.

Campaign configuration

ID	Functional Requirement	Priority
FR-CFG-001	The system shall allow marketing to configure a campaign using a landing path rather than a complete URL. The system shall combine the landing path with the registered Buyer App domain.	MUST
FR-CFG-002	The system shall configure an attribution window for each campaign.	MUST
FR-CFG-003	The system shall default the attribution window to seven days when no campaign-specific value is provided.	MUST
FR-CFG-004	The system shall allow the campaign attribution window to be edited.	MUST

FR-CFG-005	The system shall require a commission rate for every affiliate associated with a campaign.	MUST
FR-CFG-006	The system shall associate one or more Buyer Apps with a campaign.	MUST
FR-CFG-007	The system shall configure the traffic split for Buyer Apps associated with a campaign.	MUST
FR-CFG-008	The system shall prevent saving a campaign whose configured Buyer App splits do not total 100%.	MUST
FR-CFG-009	The system shall allow only eligible Buyer Apps with a signed agreement to receive a non-zero campaign share.	MUST
FR-CFG-010	The system shall prevent campaign activation until Buyer Apps, splits and landing paths are valid together.	MUST

Buyer App registry

ID	Functional Requirement	Priority
FR-CFG-011	The system shall maintain Buyer App registration information.	MUST
FR-CFG-012	The system shall maintain the Buyer App website domain.	MUST
FR-CFG-013	The system shall maintain Buyer App tier information.	MUST
FR-CFG-014	The system shall maintain Buyer App agreement status.	MUST
FR-CFG-015	The system shall maintain event schema version information for each Buyer App.	MUST
FR-CFG-016	The system shall maintain postback credentials/configuration for the Buyer App integration.	MUST
FR-CFG-017	The system shall maintain Buyer App mobile-app readiness information.	MUST
FR-CFG-018	The system shall maintain the Buyer App landing path configuration per campaign.	MUST

The PRD explicitly separates configuration into **per-Buyer-App** and **per-campaign** configuration.

Schema/version management

ID	Functional Requirement	Priority
FR-CFG-019	The system shall publish a versioned event schema for Buyer App integrations.	MUST
FR-CFG-020	Publishing a new event schema shall not require a Redbull application release.	MUST
FR-CFG-021	The system shall support Buyer App migration within the configured deprecation window.	MUST
FR-CFG-022	Adding a Buyer App shall not require a code change on the Redbull side.	MUST

Configuration versioning and rollback

ID	Functional Requirement	Priority
FR-CFG-023	The system shall version every published configuration change.	MUST
FR-CFG-024	The system shall record who changed configuration, when it was changed, and the reason for the change.	MUST
FR-CFG-025	The system shall allow a published configuration to be rolled back to an earlier version.	MUST
FR-CFG-026	A rollback shall become effective from the next click.	MUST
FR-CFG-027	A published configuration change shall apply to subsequent requests without requiring a release or restart.	MUST
FR-CFG-028	Production data shall not flow for a Buyer App until its agreement status is signed.	MUST

B. Click Routing — S2

The PRD describes S2 as receiving the affiliate click, recording it, generating the click ID, applying the traffic split, selecting the Buyer App and redirecting the shopper.

Click capture

ID	Functional Requirement	Priority
FR-CLK-001	The system shall record the affiliate click before returning a response to the shopper.	MUST
FR-CLK-002	The system shall extract `aff_click_id` from the incoming affiliate link.	MUST
FR-CLK-003	The system shall extract `sub_id` from the incoming affiliate link.	MUST
FR-CLK-004	The system shall store the affiliate's `aff_click_id` and `sub_id` against the click.	MUST
FR-CLK-005	The system shall generate the Redbull ONDC `click_id` for each click.	MUST
FR-CLK-006	The generated `click_id` shall be stored with the affiliate identifiers.	MUST
FR-CLK-007	The system shall pass the generated Redbull `click_id` to the selected Buyer App in the URL	MUST

Routing

ID	Functional Requirement	Priority
FR-CLK-008	The system shall select a Buyer App using the campaign's configured weighted traffic split.	MUST
FR-CLK-009	The selected Buyer App shall be stored against the click record.	MUST
FR-CLK-010	The system shall redirect the shopper to the selected Buyer App with appending the created ONDC click_id.	MUST
FR-CLK-011	A routing split change shall affect subsequent clicks.	SHOULD
FR-CLK-012	A click that has already been routed shall retain the Buyer App it reached.	SHOULD
FR-CLK-013	An affiliate link shall remain valid when the campaign routing split changes.	SHOULD
FR-CLK-014	If the selected Buyer App is unavailable, the routing request shall fail and the failure shall be recorded against the campaign.	MUST
FR-CLK-015	The system shall not automatically reroute to another Buyer App when the selected Buyer App is unavailable in Phase 1.	MUST

These directly correspond to the seven S2 requirements.

C. Postback and Attribution — S3

This is an important point: **S3 is not simply an attribution service.**

The PRD explicitly says S3 is the **CCTV contract that the Buyer App integrates against**, while the matching logic is also handled there.

Event contract

ID	Functional Requirement	Priority
FR-CCTV-001	The system shall expose one published event schema for Buyer App integrations.	MUST
FR-CCTV-002	CCTV shall not require per-Buyer-App payload interpretation(Common schema for all)	MUST
FR-CCTV-003	The system shall support landing events.	MUST
FR-CCTV-004	The system shall support product-view events.	MUST
FR-CCTV-005	The system shall support category-view events.	MUST
FR-CCTV-006	The system shall support add-to-cart events.	MUST
FR-CCTV-007	The system shall support checkout-start events.	MUST
FR-CCTV-008	The system shall support purchase/order-confirmation events.	MUST
FR-CCTV-009	The system shall support order-status-update events.	MUST
FR-CCTV-010	Checkout shall be represented as a separate event from cart and purchase.	MUST
FR-CCTV-011	Purchase events shall contain order ID, order value, items and item categories.	MUST

Event ingestion

ID	Functional Requirement	Priority
FR-CCTV-012	The system shall accept events containing a `click_id`.	MUST
FR-CCTV-013	The system shall accept events without a `click_id` and store them as unattributed where permitted by Buyer App tier.	MUST
FR-CCTV-014	The system shall not guess or infer a missing `click_id`.	MUST
FR-CCTV-015	Every event shall be authenticated.	MUST
FR-CCTV-016	Every event shall carry a Buyer-App-issued event ID.	MUST
FR-CCTV-017	Redelivered events shall not be counted twice.	MUST
FR-CCTV-018	Out-of-order events shall be stored once and shall not change an attribution that has already been made.	MUST
FR-CCTV-019	Tier 1 Buyer Apps shall only have events carrying a click ID accepted.	MUST
FR-CCTV-020	Tier 2 Buyer Apps shall be able to submit events without a click ID.	MUST

Attribution — S3

Click/order matching

ID	Functional Requirement	Priority
FR-ATTR-001	The system shall associate an event with the click_id carried by that event.	MUST
FR-ATTR-002	The system shall attribute an event/order only when the associated click falls within the campaign's internal attribution window.	MUST
FR-ATTR-003	The internal attribution window and affiliate contractual window shall be evaluated independently.	MUST
FR-ATTR-004	The system shall allow an order to be inside one window and outside the other.	MUST

FR-ATTR-005	An order shall never be attributed to a Buyer App that the shopper did not reach.	MUST
FR-ATTR-006	An order without a click ID shall not be attributed.	MUST
FR-ATTR-007	Unattributed orders shall contribute to the unattributed rate.	MUST

Category attribution

ID	Functional Requirement	Priority
FR-ATTR-008	The system shall calculate category match as the share of attributed order value belonging to the promoted category.	MUST
FR-ATTR-009	The system shall retain the complete category breakdown for an order rather than only the category-match percentage.	MUST
FR-ATTR-010	The retained category breakdown shall support visibility of cross-sell.	MUST

E. Reporting & Insights — S4

This is the largest functional area.

The PRD says S4 owns dashboards, funnels, baselines, the rule set and reports.

Metric/reporting

ID	Functional Requirement	Priority
FR-REP-001	The system shall define each metric once and compute it consistently across all views.	MUST
FR-REP-002	The funnel shall be filterable/sliceable by affiliate.	MUST
FR-REP-003	The funnel shall be filterable/sliceable by Buyer App.	MUST
FR-REP-004	Affiliate and Buyer App funnel slices shall reconcile to the same totals.	MUST
FR-REP-005	Reports shall aggregate to category.	MUST

FR-REP-006	SKU, seller and provider data shall not cross between Buyer Apps in reports.	MUST
FR-REP-007	Buyer App reports shall not identify competitors by name.	MUST
FR-REP-008	Buyer App benchmark reports shall require a cohort of at least five Buyer Apps.	MUST
FR-REP-009	The recipient shall be excluded from the median shown to that recipient.	MUST
FR-REP-010	Buyer App cohort membership shall not be disclosed.	MUST

Daily affiliate view

ID	Functional Requirement	Priority
FR-REP-011	The daily view shall show orders today and yesterday per affiliate.	MUST
FR-REP-012	The daily view shall show day-on-day order change.	MUST
FR-REP-013	The daily view shall show month-to-date pacing against target.	MUST
FR-REP-014	The daily view shall show month-end run-rate projection.	MUST
FR-REP-015	The daily view shall show GMV.	MUST
FR-REP-016	The daily view shall show AOV.	MUST
FR-REP-017	The daily view shall show commission payable.	MUST
FR-REP-018	The daily view shall show completed versus cancelled/returned share.	MUST
FR-REP-019	The daily view shall show Buyer App and category splits.	MUST
FR-REP-020	Where affiliate-claimed figures exist, the daily view shall show them alongside platform-computed figures.	SHOULD

FR-REP-021	The system shall show the difference between affiliate-claimed and platform-computed transactions and values.	SHOULD
FR-REP-022	Concentration-cap and pacing violations shall be surfaced in the daily view.	MUST
FR-REP-023	The daily view shall retain sufficient history for week-on-week and month-on-month patterns.	SHOULD

Insights / decision queue

ID	Functional Requirement	Priority
FR-INS-001	The system shall evaluate the defined deterministic rule set against attributed data.	MUST
FR-INS-002	The system shall generate a flag when a rule condition is satisfied.	MUST
FR-INS-003	Each flag shall identify the rule that generated it.	MUST
FR-INS-004	Each flag shall expose its condition, observed values and threshold.	MUST
FR-INS-005	Flags shall be ranked by estimated value at stake.	MUST
FR-INS-006	Marketing users shall be able to edit rule thresholds.	MUST
FR-INS-007	Rule logic shall not be editable in the Phase 1 product.	MUST
FR-INS-008	Dismissal rate shall be tracked per rule.	MUST
FR-INS-009	Dismissal rate shall be visible.	MUST
FR-INS-010	A user shall be able to dismiss a flag with a reason.	SHOULD
FR-INS-011	Dismissal reasons shall be retained.	SHOULD
FR-INS-012	R-14 through R-17 shall use the same daily affiliate figures already computed by reporting.	MUST

FR-INS-013	If affiliate-claimed data required by R-16 is unavailable, the rule shall not fire.	MUST
FR-INS-014	The affiliate shall instead be marked reconciliation unavailable.	MUST

Affiliate reports

ID	Functional Requirement	Priority
FR-REP-024	The system shall generate a report for any campaign, affiliate and period.	MUST
FR-REP-025	The report shall include the affiliate's aff_click_id and sub_id.	MUST
FR-REP-026	A conversion shall appear once in a report period.	MUST
FR-REP-027	Re-running a report for the same period shall return the same conversions.	MUST
FR-REP-028	The affiliate contractual attribution window shall determine which conversions appear in an affiliate report.	MUST

F. UI — S5

The PRD explicitly defines S5 as the browser interface for configuration, reading dashboards and sending reports.

ID	Functional Requirement	Priority
FR-UI-001	Administrators shall be able to create and modify configuration through the UI.	MUST
FR-UI-002	General users shall have read access to configuration.	MUST
FR-UI-003	The UI shall display configuration versions and audit history.	MUST
FR-UI-004	The UI shall allow an earlier configuration version to be rolled back.	MUST

FR-UI-005	The UI shall display dashboards, funnels and the decision queue.	MUST
FR-UI-006	The UI shall support producing, drafting and sending affiliate reports.	SHOULD
FR-UI-007	The platform shall compose the subject and body of affiliate report emails.	MUST
FR-UI-008	Marketing shall be able to edit the subject and body before sending.	MUST
FR-UI-009	Every affiliate report send shall be recorded with campaign, affiliate, period, recipients, sender and timestamp.	MUST
FR-UI-010	The record of each send shall be visible in the product.	MUST

The report/email behavior is explicitly specified in the S5 requirements.

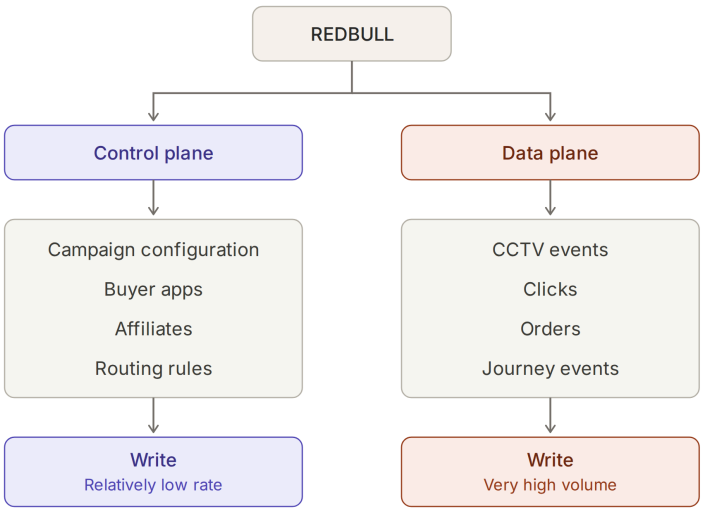
3.2 Non Functional Requirements

Quality	**Requirement**
Redirect latency	Under 300 ms p95, excluding the Buyer App's own page load
Click durability	Every click recorded before any response, abandoned ones included
Freshness	Data no more than 120 min behind live
Dashboard speed	A view loads in under 2 s
Counting integrity	Redelivered events and repeated orders counted once
Consistency	One definition per metric; slices reconcile to the same totals
Determinism	Same data gives the same answer; a person can reproduce every flag
Report stability	Re-running a period returns the same conversions
Change without release	Config and schema changes live from the next request, no restart; a new Buyer App needs no code
Auditability	Who, when and why for every config change; every send recorded; dismissal reasons kept
Security	Every event authenticated; only admins change configuration

Privacy	Signed agreement before any traffic or data; Tier 2 off by default; cohort of 5 or more; category level only; no shopper-level data either way
Outcome targets	Over 80% of orders on tagged Buyer Apps attributed; over 95% of test events pass before go-live; no rule above 50% dismissal without review
Availability	Not specified
Throughput and scale	Not specified
Alert timeliness	"Within minutes" implied for the outage response
Retention	Not specified

4. Architecture Style

The platform follows a **service-based architecture with event-driven integration**. And one more thing is we are going to design the write heavy application.



This was chosen because the platform has two different types of workloads:

- **Real-time operations** — click routing and Buyer App event ingestion must be fast and reliable.
- **Background processing** — attribution, reporting, and analytics can happen asynchronously and do not need to block real-time requests.

To support this, the platform is divided into **five services**, with each service owning its own data. There is no shared database, which keeps the real-time click flow independent from reporting and analytics workloads.

Communication follows a simple rule:

- **Synchronous calls** are used when an immediate response is required, such as shopper redirection or Buyer App requests.
- **Durable events** are used when processing can happen asynchronously, such as click recording, attribution, and reporting.

Configuration is distributed as **versioned snapshots**, allowing services to continue operating even when the Configuration service is temporarily unavailable.

Reporting and analytics are processed periodically into read models rather than being part of the real-time flow. This keeps the operational path fast while still meeting the required reporting freshness.

Why This Architecture

This approach provides:

- **Low latency** for real-time operations.
- **Reliability** through durable event processing.
- **Failure isolation** between operational and analytical workloads.
- **Independent data ownership** for each service.
- **Simpler operations** than a highly fragmented microservice architecture.

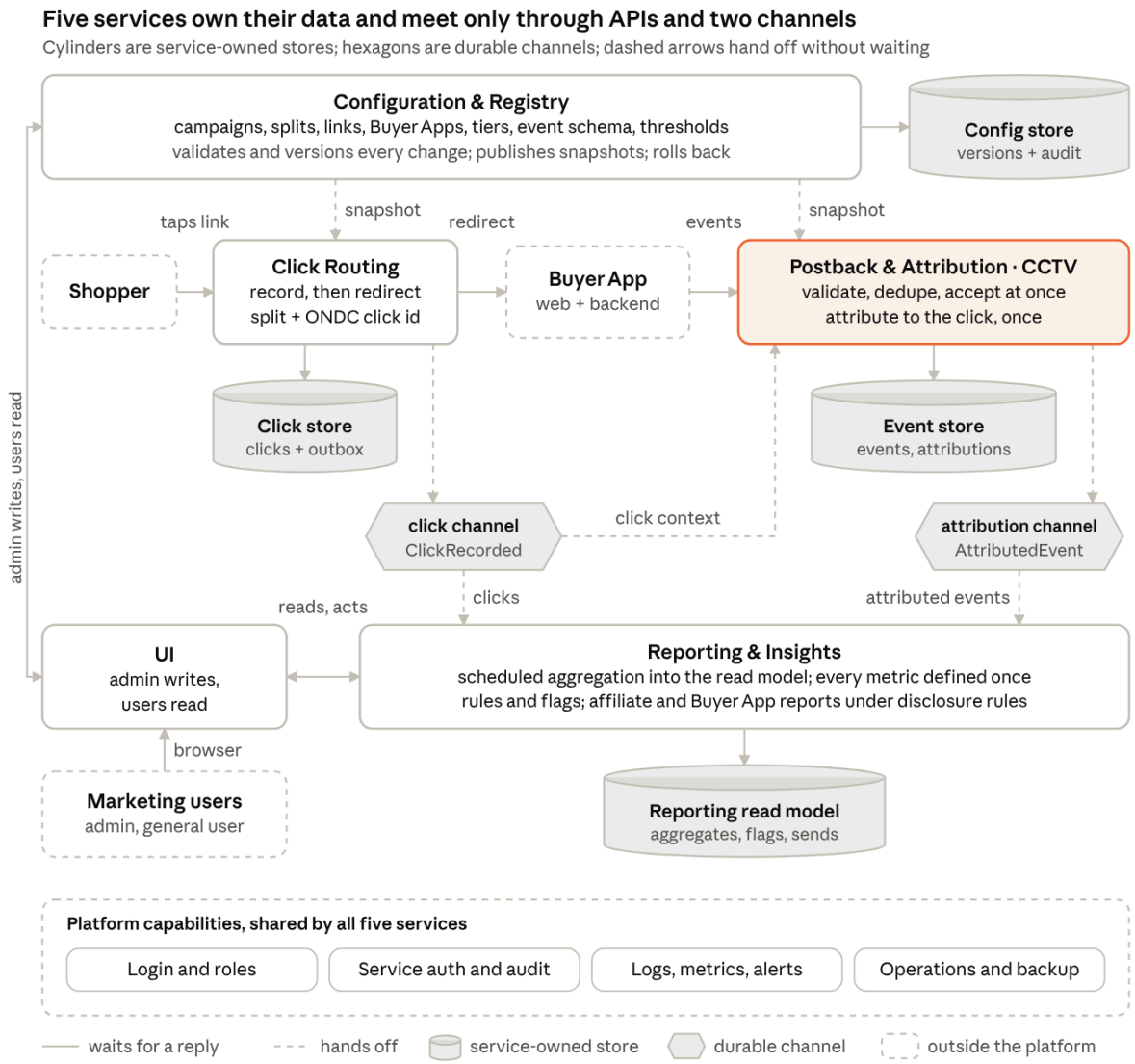
We deliberately use a **small number of services** rather than many fine-grained microservices because further splitting would add network and operational complexity without providing significant additional isolation at the current scale.

5. Architecture

Apart from the NFR's specified in the PRD, we have made few assumptions to come up with this architecture

Few Assumptions

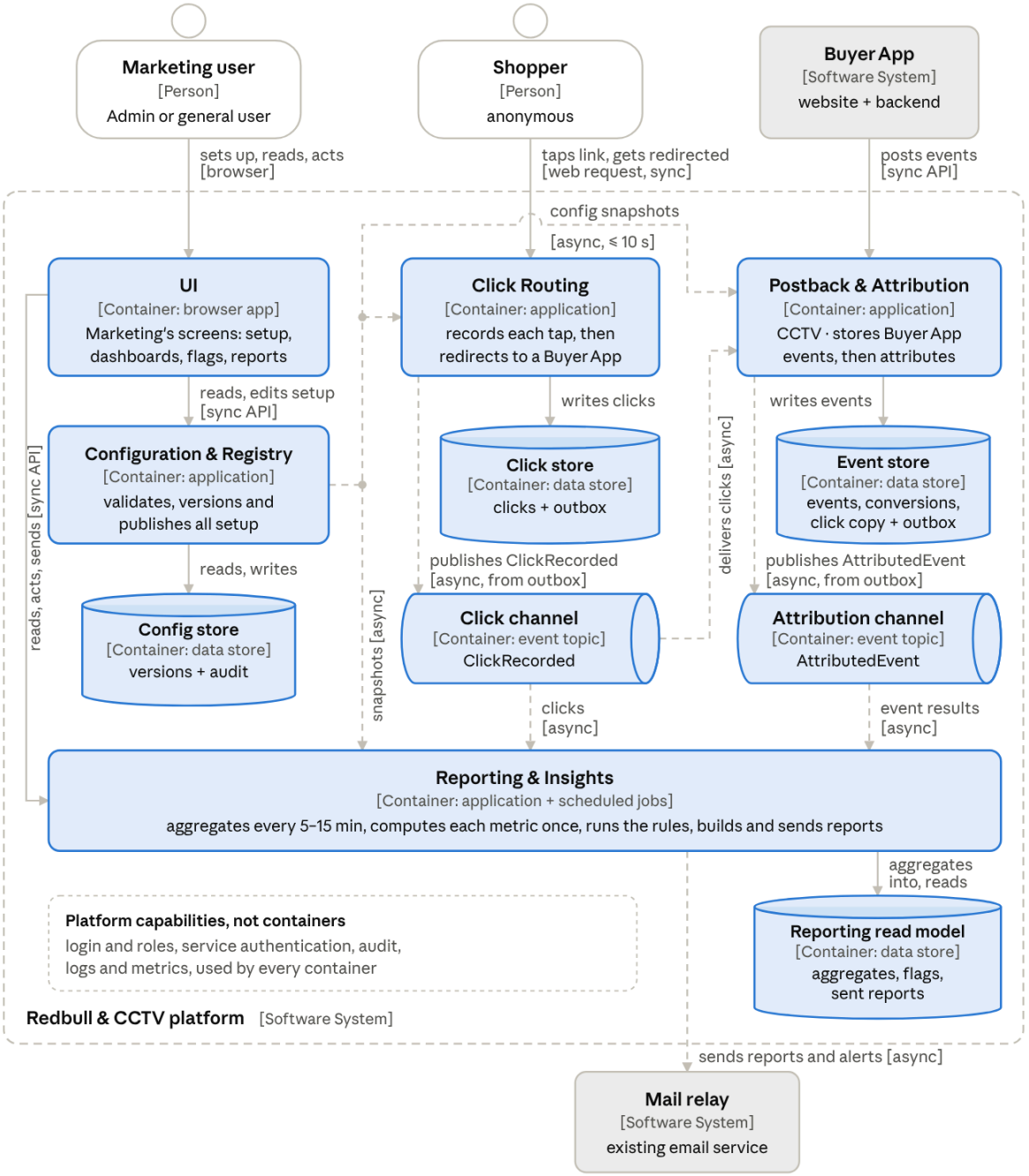
- 500,000 clicks a day, with bursts of 200 a second.
- 10 million Buyer App events a day, peaking at 600 a second. 15,000 attributed orders a day.
- Uptime of 99.9% for Click Routing and Postback, 99.5% for the other services.
- 13 months of detail kept.



The below architecture is the C2 model style of architecture.

Containers: five applications own their data and meet through two channels

C4 level 2 · scope: the Redbull & CCTV platform · technology still to be chosen



Not drawn, to keep the lines clear: the UI calling Postback & Attribution for a Buyer App's test results and to check sample events (synchronous).

person · external · application · data store · event topic · sync, waits · async, hands off

Container	Responsibility	Data it owns
UI	Marketing's screens for setup, dashboards, flags, reports, claimed-figure uploads and test results; no business logic	
Configuration & Registry	Rulebook: Validates, versions and publishes all setup; rolls back by republishing an earlier version	Config Store
Click Routing	Resolves the campaign link, checks it is live, draws a Buyer App by weight, creates the ONDC click id, records the click, then redirects	Click store
Postback & Attribution (CCTV)	Authenticates the Buyer App; checks agreement, tier and schema; stores the event and answers; attributes it in the background	Event Store
Reporting & Insights	Aggregates every 5–15 min, computes each metric once, runs the rules, builds and sends reports under the disclosure rules	Reporting Read Model
Config store	Campaigns, splits, links, Buyer Apps, affiliates, thresholds, targets, schema versions, audit	
Click store	Clicks keyed by ONDC click id, plus the outbox	
Event store	Events (Buyer App + event id), click copy, conversions (Buyer App + order id), outbox	
Reporting read model	Aggregates, baselines and cohorts, rebuildable from the channels; plus records only Reporting holds: rule evaluations and flags with their dismissals, claimed figures and sent reports	
Click channel	ClickRecorded for every recorded click, failed routings included	
Attribution channel	AttributedEvent for every stored event, with its result	

RelationShips

From → to	What	Style
Marketing user → UI	Sets up, reads dashboards, acts on flags, uploads claimed figures, sends reports	Sync, browser
Shopper → Click Routing	Taps the campaign link; gets the redirect with the ONDC click id	Sync, web request
Buyer App → Postback & Attribution	Posts funnel and order events; gets accept or reject with a reason	Sync API
UI → Configuration & Registry	Reads and edits setup; only Admins may change it	Sync API
UI → Reporting & Insights	Reads dashboards and flags, dismisses flags, uploads claimed figures, sends reports	Sync API
Configuration & Registry → Click Routing, Postback & Attribution, Reporting & Insights	Publishes immutable configuration snapshots	Async, within 10 s
Each application → its own store	Reads and writes; the click is written before the redirect	Sync
Click Routing → click channel	Publishes ClickRecorded from its outbox	Async
Click channel → Postback & Attribution, Reporting & Insights	Delivers every click, at least once	Async

Postback & Attribution → attribution channel	Publishes AttributedEvent from its outbox	Async
Attribution channel → Reporting & Insights	Delivers every event with its result, at least once	Async
Reporting & Insights → mail relay	Hands over partner reports and outage alerts	Async

4. Every piece of data, from creation to use

Every piece of data has one creator and one owning store; every other service works from a copy that arrives along a drawn line. Only two things change: configuration, which changes by adding versions, and figures, which each run recomputes from the facts.

Data	**Created by**	**Owned by**	**Read by, and how it gets there**
Configuration: Buyer Apps, campaigns, splits, landing paths, windows, commission, thresholds, targets	An Admin's save in the UI, checked as a whole by Configuration & Registry	Config store	Click Routing, Postback & Attribution and Reporting & Insights, each from its own snapshot, pushed within 10 s. The UI through Configuration's API
Event schema	Configuration & Registry, published once	Config store	Postback & Attribution's checker, through the snapshot. Each Buyer App builds against it once, at onboarding

Campaign link, one per campaign per affiliate	Configuration & Registry, when the campaign is activated	Config store	The affiliate, who receives it from Marketing. Click Routing, which resolves it from its snapshot
The affiliate's aff_click_id and sub_id	The affiliate, added to the link	Click store, on the click, exactly as received	Postback & Attribution and Reporting & Insights, through the click channel. The affiliate gets them back in its report
Click record	Click Routing, on every tap, before the redirect	Click store	Postback & Attribution, which keeps a click copy, and Reporting & Insights, both through the click channel
ONDC click id	Click Routing, one per click	Click store, as the click's key	The Buyer App, through the redirect; its own page keeps it on its own domain and sends it back with every event
Event: landing, product view, category view, add to cart, checkout started	The Buyer App's backend, with its own event id	Event store, once per Buyer App + event id, stored before the answer	The Attributor in the same container. Reporting & Insights, with the event's result, through the attribution channel

Order: the order_confirmed event	The Buyer App's backend	Event store, once per Buyer App + order id, with value, items and categories	The Attributor. Reporting & Insights, through the attribution channel
Event result: attributed or not, with the reason	The Attributor, in the background, once per stored event	Event store	Reporting & Insights, through the attribution channel
Conversion	The Attributor, when an order matches a click from the same Buyer App inside at least one window	Event store, at most one per Buyer App + order id, with its report period fixed then	Reporting & Insights, through the attribution channel, for figures and affiliate reports
Order status update: cancelled or returned	The Buyer App's backend	Event store	Reporting & Insights, through the attribution channel, for the completed against cancelled or returned share
Figures: funnels, conversions, category match, GMV, commission, ROI, baselines, the daily affiliate view	Reporting & Insights, every 5–15 minutes	Reporting read model	The UI, through Reporting's API. The rules and the reports, inside Reporting & Insights

Flag and its dismissal	Reporting & Insights' rules, after each run, with inputs and threshold version stored	Reporting read model	The UI's decision queue, ranked by value at stake
Report and its send record	Reporting & Insights, when Marketing builds and sends a report in the UI	Reporting read model	The affiliate or Buyer App, by email through the mail relay. Marketing, in the send history

Users, roles and sessions belong to the shared login capability, not to any of the five services. Each audit record stays with the service that owns the action

5. Architectural Design Records(ADR)

ADR-001 — Use Five Independently Deployable Services

Status: Proposed

Context

The system has five distinct responsibilities:

- Configuration & Registry
- Click Routing
- Postback & Attribution
- Reporting & Insights
- UI

They have different workload characteristics and failure boundaries. Click Routing has a strict redirect-latency requirement, while Reporting performs heavier processing and can tolerate delayed data. The PRD also explicitly structures the system around these five services.

Decision

Implement the platform as five independently deployable services:

1. Configuration & Registry

2. Click Routing
3. Postback & Attribution
4. Reporting & Insights
5. UI

Alternatives considered

- **Single application:** rejected because reporting workloads and releases could affect the click path.
- **More fine-grained services:** rejected because the additional boundaries would add complexity without a clear requirement for further isolation at the expected scale.

Consequences

Benefits

- Independent scaling and deployment.
- Clear ownership boundaries.
- Failure isolation between major workloads.

Costs

- More services to deploy, monitor and secure.
- Communication between services must be explicitly designed.

ADR-002 — Keep Campaign and Buyer App Configuration Together

Status: Proposed

Context

Campaign configuration depends on Buyer App information such as domain, tier and agreement status. A campaign cannot be considered valid without this information.

Decision

Configuration & Registry owns both:

- Buyer App registry
- Campaign configuration

and versions them together.

Alternatives considered

Separate Buyer App Registry and Campaign Configuration services.

Consequences

A campaign can be validated against the relevant Buyer App information within one ownership boundary, and configuration versions and rollback remain consistent.

ADR-003 — Each Service Owns Its Data

Status: Proposed

Context

Different services have different access patterns. Click Routing requires fast transactional access, while Reporting processes large amounts of historical data.

A shared database would make the same data accessible to multiple services and couple their schemas.

Decision

Each service owns the data required for its responsibility and other services do not directly access its database.

The current design has separate stores for:

- Configuration
- Clicks
- Events
- Reporting/read-model data

Alternatives considered

A shared database used by multiple services.

Consequences

Benefits

- Clear data ownership.
- Independent schema evolution.
- Independent scaling and optimization.

Costs

- Data required by another service must be transferred or replicated.
- Cross-service consistency must be explicitly designed.

The source ADR correctly identifies this as a major architectural boundary.

ADR-004 — Distribute Configuration as Versioned Snapshots

Status: Proposed

Context

Click Routing needs configuration during every click. Calling Configuration & Registry synchronously for every click would add a network dependency to the redirect path.

Decision

Configuration & Registry publishes **immutable, versioned configuration snapshots**.

S2, S3 and S4 maintain their own current copy of the configuration they need.

Each click records the configuration version used for routing.

Rollback publishes an earlier configuration version.

Alternatives considered

- Read Configuration & Registry on every request.
- Read Configuration & Registry's database directly.

Both introduce runtime coupling to S1.

Consequences

Benefits

- Click routing does not depend on S1 being available for every request.
- Configuration can be reproduced using the recorded version.
- Rollback is explicit.

Cost

There can be a short propagation delay between publishing a configuration and all consumers receiving it

ADR-005 — Support Out-of-Order Click and Event Arrival

Status: Proposed

Context

Clicks and Buyer App events arrive through different paths. Therefore an event can reach S3 before the corresponding click has arrived.

Immediately declaring such an event unattributed could incorrectly lose a valid attribution.

Decision

When an event references a click that has not yet arrived:

```
None
Event received ->Store event ->Waiting for click->Click arrives ->Attribution
```

The event is marked as unknown only when the system has sufficient evidence that the referenced click does not exist.

Alternatives considered

- Reject the event.

- Immediately mark it unattributed.
- Synchronously query S2's database.
- Use a fixed timeout.

Consequences

Benefits

- Protects attribution correctness.
- Keeps service data ownership boundaries intact.

Costs

- Requires a waiting/reconciliation mechanism.
- Requires monitoring of waiting-event age.

ADR-006 — Use Asynchronous Channels for Cross-Service Data Propagation

Status: Proposed

Context

S3 requires click information from S2, while S4 requires clicks and attribution results. Direct service calls or database reads would couple these services.

Decision

Use asynchronous data channels:

```
None
S2 ── Click Channel ──> S3

S2 ── Click Channel ──> S4

S3 ── Attribution Channel ──> S4
```

Consumers maintain their own copies of the data required for their responsibilities.

Alternatives considered

- Direct database access.
- Synchronous lookup for every event.
- Fully shared event bus with no explicit ownership of channels.

Consequences

Benefits

- Producers do not wait for consumers.
- Consumers can catch up after temporary failures.
- Services remain independently deployable.

Costs

- Eventual consistency.
- Additional infrastructure for delivery and replay.

ADR-007 — Use Scheduled Aggregation for Reporting

Status: Proposed

Context

S4 needs:

- consistent metrics,
- dashboards under two seconds,
- data freshness within 120 minutes.

The requirements do not require second-level real-time reporting.

Decision

S4 maintains its own reporting/read model and periodically computes the reporting metrics and rules from the received data.

The UI reads the prepared reporting model rather than calculating metrics dynamically.

Alternatives considered

- Calculate metrics on every dashboard request.
- Build a real-time streaming calculation pipeline.

Consequences

Benefits

- Fast and consistent dashboard reads.
- Reusable metric calculations.
- Deterministic reporting.

Costs

- Reporting is eventually consistent.
- Every view must expose its data freshness/as-of time.

ADR-008 — Keep the UI Thin and Enforce Rules at the Owning Services

Status: Proposed

Context

The same business operation can be reached through the UI or other service interfaces. Putting validation and authorization only in the UI would allow those rules to be bypassed.

Decision

S5 is responsible for presentation and user interaction.

Business validation, authorization and data ownership remain with the services responsible for that business capability.

Consequences

Benefits

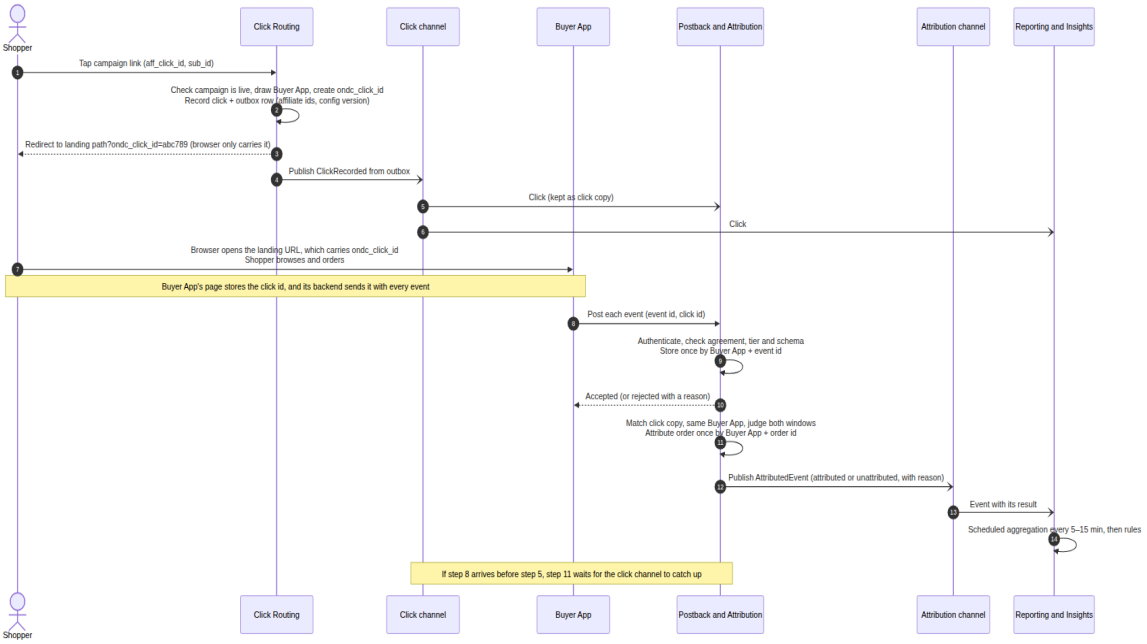
- Business rules have one authoritative location.
- Non-UI access is protected.
- UI changes do not require duplicating business logic.

Cost

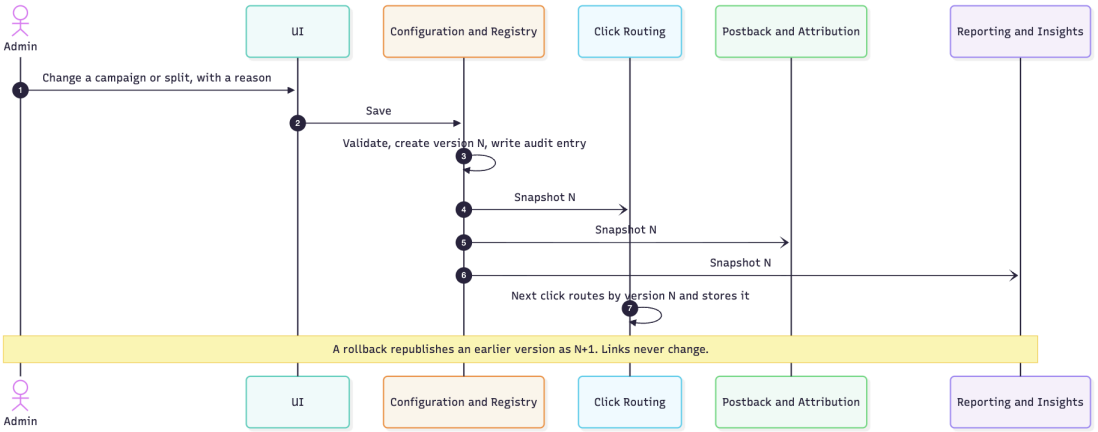
The UI may need to coordinate calls across multiple services.

7. Key flows through the design

Click to Attributed Order



Configuration change and rollback



Decisions and partner reports

